

01 Sequence Limits (textbook 5.1)

Does the sequence have a limit as $n \rightarrow \infty$? If yes, find it.

a. $a_n = \frac{3-n}{2}$

b. $a_n = \frac{\sqrt[4]{n}}{n^3}$

c. $a_n = \frac{n-1}{n^2-1}$

d. $a_n = 1^n$

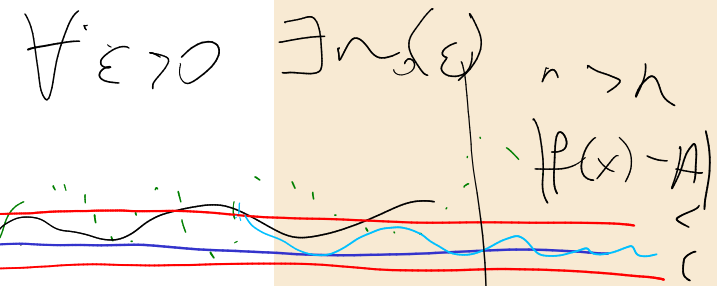
e. $a_n = 0.4^n$

f. $a_n = (-4)^n$

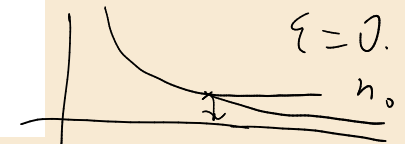
g. Pick any number as a_1 , then $a_{n+1} = a_n/1.5$.

h. $a_1 = 1$; for $n \geq 2$, let a_n be the first digit after the decimal point of $a_{n-1}/7$.

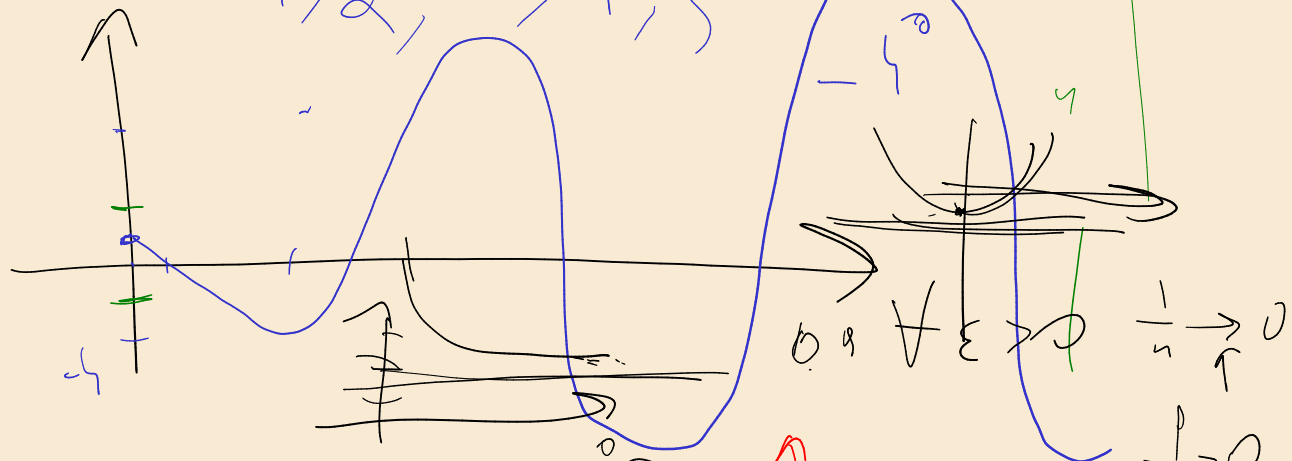
Hint: write out the first few terms; for the last one, a calculator helps.



$f(x) = 5$
 $\epsilon = 1$
 $\epsilon = 0.5$



1, 2, 3, 4, 5



06a, b ✓

01 z ✓

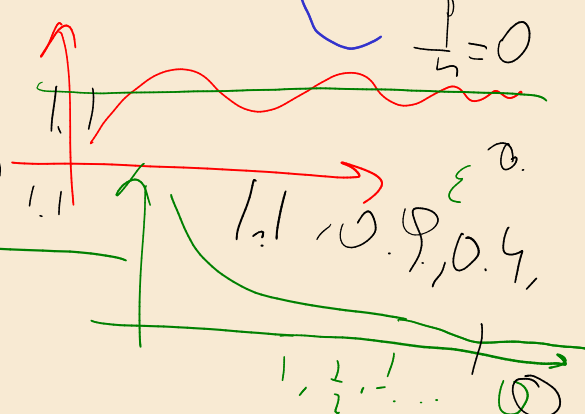
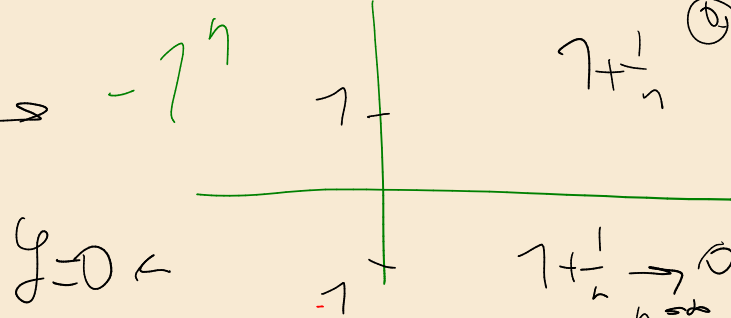
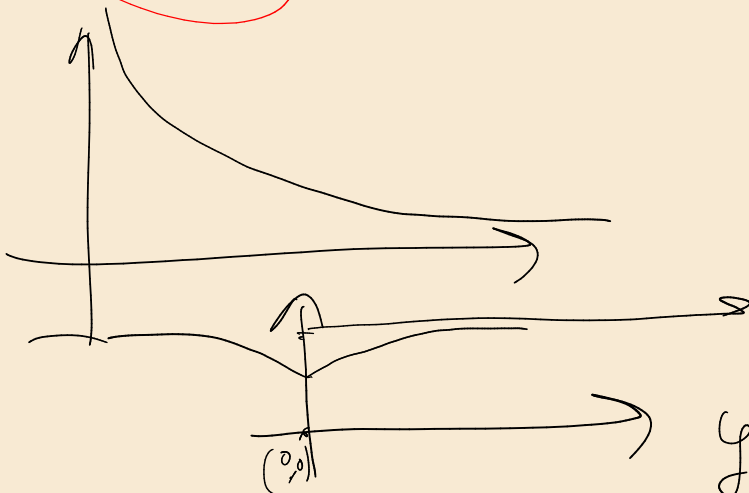
2h\$kr\$gh\$u\$

03 d ✓

0.9

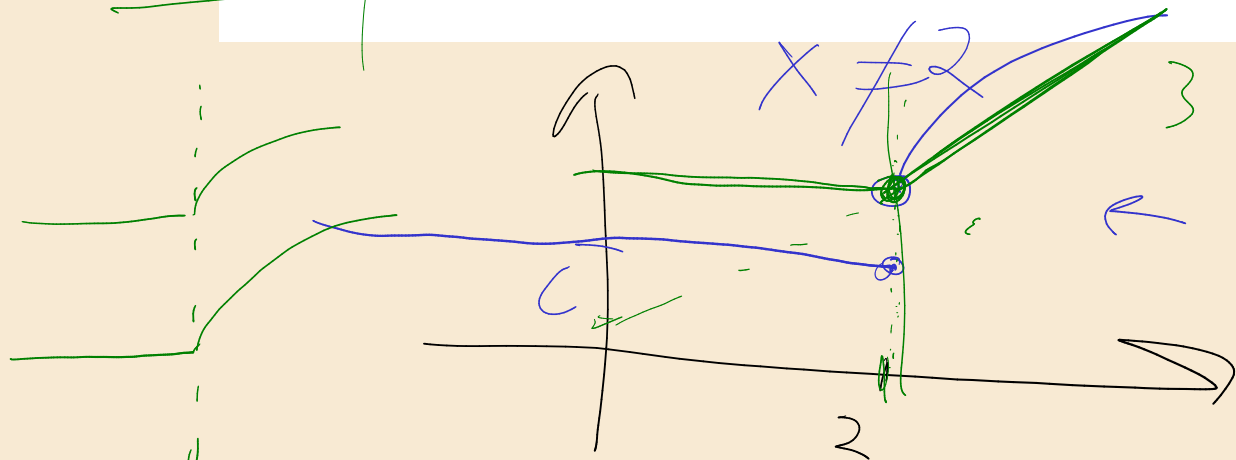
05.6 ~ x^x

84



Խնդիր 5.3 Ներկայալ ֆունկցիաները բաղկացած են $x = 2$ կետում իրար «սոսնձած» երկու առանձին ֆունկցիաներից: Կարո՞ղ եք գտնել c -ի այնպիսի արժեք, որի դեպքում ֆունկցիան անընդհատ լինի 2-ում:

$$\eta. f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{երբ } x \geq 2 \\ c & \text{երբ } x < 2 \end{cases}$$



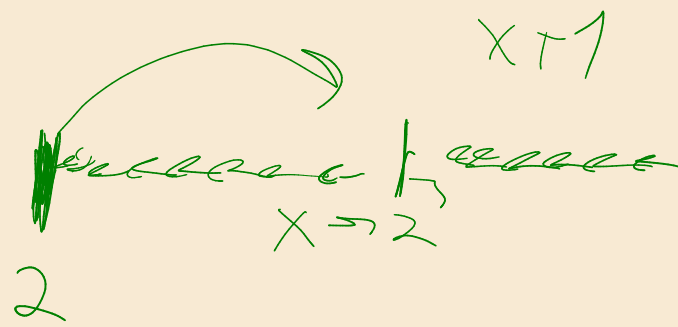
$$\lim_{x \rightarrow 2^+} \frac{x^2 - x - 2}{x - 2}$$

$$\frac{x^2 - x - 2}{x - 2} \xrightarrow{x \rightarrow 2^+}$$

$$\frac{(x-2)(x+1)}{x-2}$$

$$\xrightarrow{x \rightarrow 2} x+1 \rightarrow 3$$

3.5 2 2.000!
3.1 3.01 → 3.0001



$$X \rightarrow 2$$

06 ML Activation Derivatives (textbook 5.6)

Two functions used heavily in machine learning:

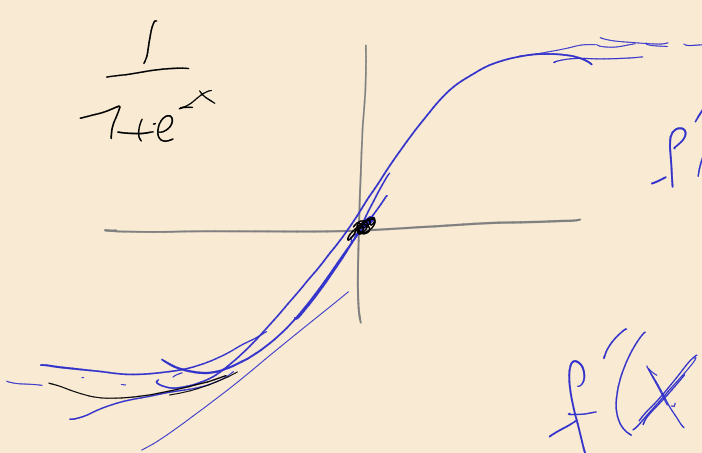
a. **ReLU** (rectified linear unit):

$$f(x) = \begin{cases} x & x > 0 \\ 0 & x \leq 0 \end{cases}$$

b. **Sigmoid**:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Compute their derivatives. Can you express the sigmoid's derivative in terms of σ itself?



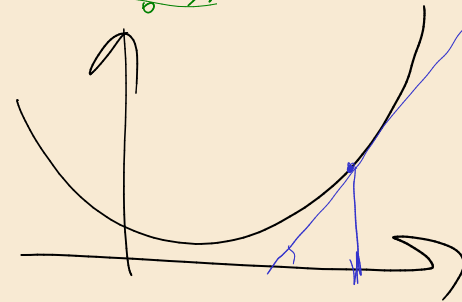
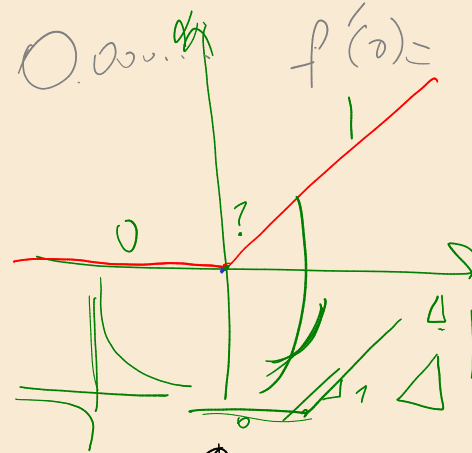
$$f'(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \\ ?, & x = 0 \end{cases}$$

$$f'(x)$$

$$\frac{1}{1+e^{-x}} + \frac{1}{1+e^{-x}} = 1$$

$$0.25 = 0.25$$

$$0.25 = 0.25$$

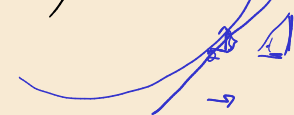


$$f(x) = x$$

$$x \rightarrow 1$$

$$x^2$$

$$\frac{x}{x} = 1$$



$$\left(\frac{1}{1+e^{-x}} \right)'$$

$$(1+e^{-x})'$$

$$\frac{1}{1+e^{-x}}$$

$$\frac{uv - vu}{u^2} = \frac{f(g(x))}{f(g(x))} = 1$$

$$x^n \rightarrow nx^{n-1}$$

$$(e^{-x})' = -e^{-x}$$

$$x^n e^x = e^x \ln x$$

$$(f+g)' = f' + g'$$

$$(e^{-x})' = (e^{-x}) \cdot (-x)$$

$$(e^x)' = e^x \cdot (x)' = e^x$$

$$(e^{-x})' = -e^{-x} \cdot 1$$

$$e^x \ln e = e^x \cdot 1 = e^x$$

$$-1(1+e^{-x})^{-2} \cdot (1+e^{-x})'$$

$$f(g(x))' = f'(g(x)) \cdot g'(x)$$

$$(2 \sin x) \cdot (\sin x)'$$

$$\sin(x) = \cos(x)$$

$$x^2 = 2x$$

$$f(g(x))' = f'(g(x)) \cdot g'(x)$$

$$e^x \ln x$$

$$1, 2, 3$$

$$2^3 = 8$$

$$\emptyset \quad \underline{2}^{\frac{1}{3}} = 8$$

$$\ln \underline{2} \underline{8}$$

$$\log_2 8 = 3$$

$$x^x$$

$$a^{\underline{x}} \quad \underline{x}^a$$

$$x \rightarrow x^{x-1}$$

$$x' \cdot x^{x-1} + x \cdot (x^{x-1})'$$

lin

$$x \cdot x^{x-2}$$

$$x^{-200} \quad x \rightarrow 0 \quad -1 - \infty$$

$$e^{\ln(x^x)}$$

$$\frac{x \ln x}{x}$$

$$\ln y = \ln x^x$$

$$y = e^x$$

$$f(x) f'(x)$$

$$a = e^{\ln a}$$

$$* \cdot \frac{1}{x}$$

$$e^{x \ln x}$$

$$\ln y = \ln x^x$$

$$x^x = e^{x \ln x}$$

$$\frac{e^{x \ln x} (1 + \ln x)}{x^x (1 + \ln x)}$$

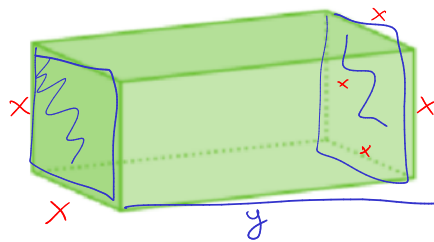


$$f(f^{-1}(x)) = x$$

$$= e^{x \ln x} \cdot (x \ln x)' = \frac{x \cdot (\ln x)' (1 + \ln x)}{(1 + \ln x)}$$

$$\begin{aligned} & e^{x \ln x} \cdot (x \ln x)' = \\ & = e^{x \ln x} (1 + \ln x) = \\ & = x^x (1 + \ln x) \end{aligned}$$

Խնդիր 5.9 Ունենք 24 մ^2 մակերեսով սրվարաթուղթ (ինչպես նաև մկրար ու սոսինձ), որով ցանկանում ենք պարարասել այսպիսի փուփ:



$$x^2 + x^2 + 4xy = 24$$

$$y = \frac{24 - 2x^2}{4x} = \frac{12 - x^2}{2x}$$

$$x > 2$$

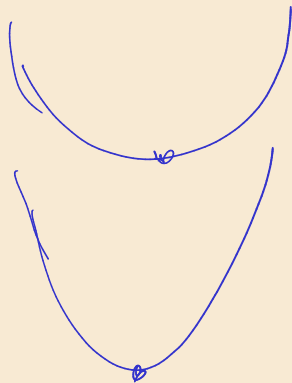
$$y = \frac{24 - 8}{8} = 2$$

որի ձախ և աջ նիստերը բառակուսիներ են:

Ամենաշարը որքա՞ն կարող է լինել այդ փուփի ծավալը:

$$V = x^2 y$$

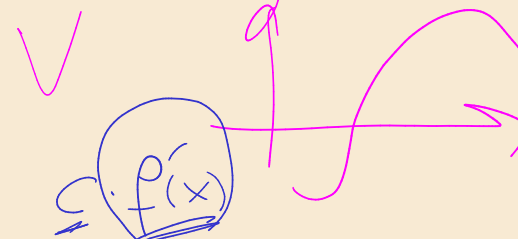
$$V = x^2 \cdot \frac{12 - x^2}{2x} = \frac{x(12 - x^2)}{2}$$



$$3 + x^2$$

$$(3 + x^2) \cdot 15$$

$$(c \cdot f(x))' = c \cdot f'(x) = 0$$

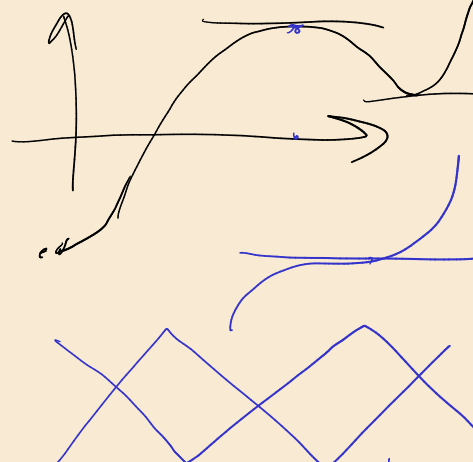
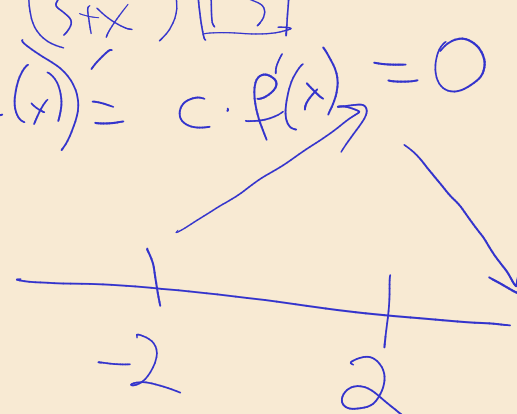


$$4x$$

$$x$$

$$16 < 5$$

$$x \cdot 0 = 0$$

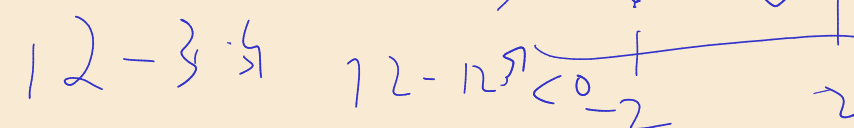


$$f'(x) = \frac{x(12 - x^2)}{2}$$

$$(12x - x^3)'$$

$$12 - 3x^2 = 0$$

$$x = \pm 2$$



$$12 - 3x^2 = 0$$

$$x = \pm 2$$

$$\frac{x^3}{e^x}$$

$$2\pi f \{$$

$$\mathbb{R}^n \rightarrow \mathbb{R}$$

$$\mathbb{R}^n \rightarrow \mathbb{R}^m$$

jacob